DaVinci 200 Safety Test

Test Description and Checklist

Document Management

Document-History

Version	Status	Date	Responsible Person	Reason for Change
1.0	Draft	13.06.2024	CG	Version erstellt auf Basis des DaVinci 200 SMIF Safety Check V1.1
1.1	Release	18.09.2024	HL	Convert to English version
1.2	Release	8.12.2024	EL	“Stage stops” added. Main Picture changed.
1.3	Release	02.10.2025	EL	Removed “Stage stops”, added OC-Loadporttest for LP2. Point 3.4

Table of Contents

1 Introduction

1.1 Purpose of Document

This document describes the positions of the safety-relevant switches. It also contains a checklist which is processed and signed by the responsible employee as part of the safety tests before delivery. This signed checklist is made available for filing together with the safety assessment of the QA.

.

1.2 Definitions and Abbreviations

Abbrevation:	Description
ARGOS	Product name of MT1300 ARGOS, 300mm Frame Inspection System
Rembrandt	Product name of MT8300 Rembrandt, 200mm/300mm Wafer Imaging System
DaVinci	Product name of a Microscope based Metrology and Inspection System
V	Voltage in Volt
A	Current in Ampere
mm	Millimeter
h	Time in Hour
d	Time in Day
EFEM	Equipment Frontend Module
MP	Measurement platform
FOUP	Front Opening Unified Pod
Frame	380mm carrier with wafer on the foil
EPLAN	Software for creating electrical plans

2 Emergency Stop Buttons

The correct function of the emergency stop buttons is checked on the running system.

Action: Press emergency stop button.

Reaction: Emergency stop is executed:

- Robot handling is stopped
- Stage stops

Documentation: Robot stops immediately → to be documented by checking “OK”. Otherwise “NOK”

Stage stops immediately → to be documented by checking “OK”. Otherwise “NOK”

Location of the emergency stop buttons on the housing: See Figure 1.

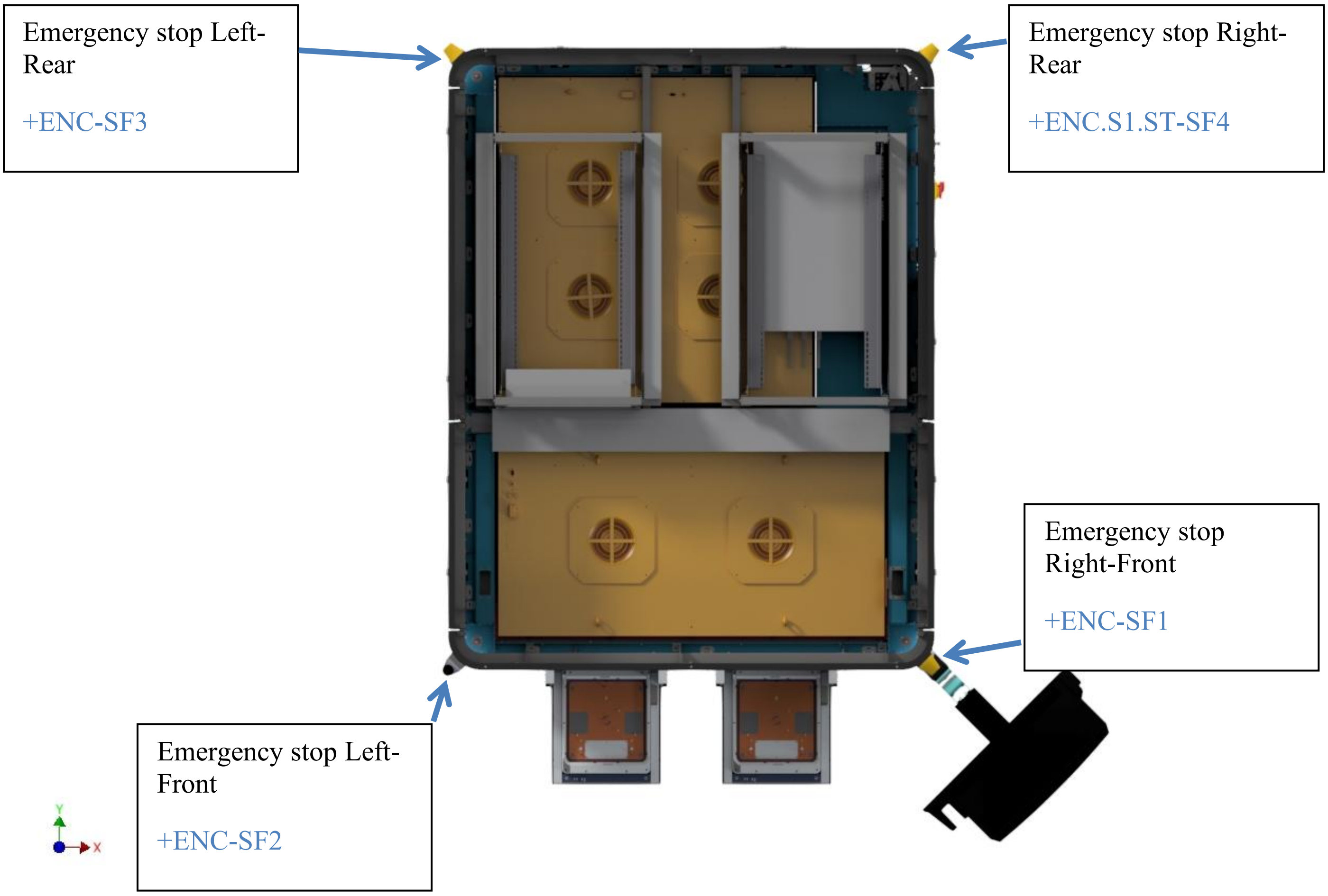


Figure 1: DAVINCI – Location of the emergency stop buttons

Emergency stop #	EPLAN	Description	Action	Rob stops
N1	+ENC-SF1	Emergency stop right front	Press emergency stop button	
N2	+ENC-SF2	Emergency stop left front	Press emergency stop button	
N3	+ENC.S1.ST-SF4	Emergency stop right rear	Press emergency stop button	
N4	+ENC-SF3	Emergency stop left rear	Press emergency stop button	
Teach-Pendant		Emergency stop Robot Teach Panel	Press emergency stop button	

3 Robot and Scanning-Stage Door and Loadport Interlocks

The correct function of all interlocks is checked on the running system.

Action: Remove the loaded cassettes during robot handling or open the control cabinet door (to the interior) during robot operation.

Reaction: Emergency stop is executed:

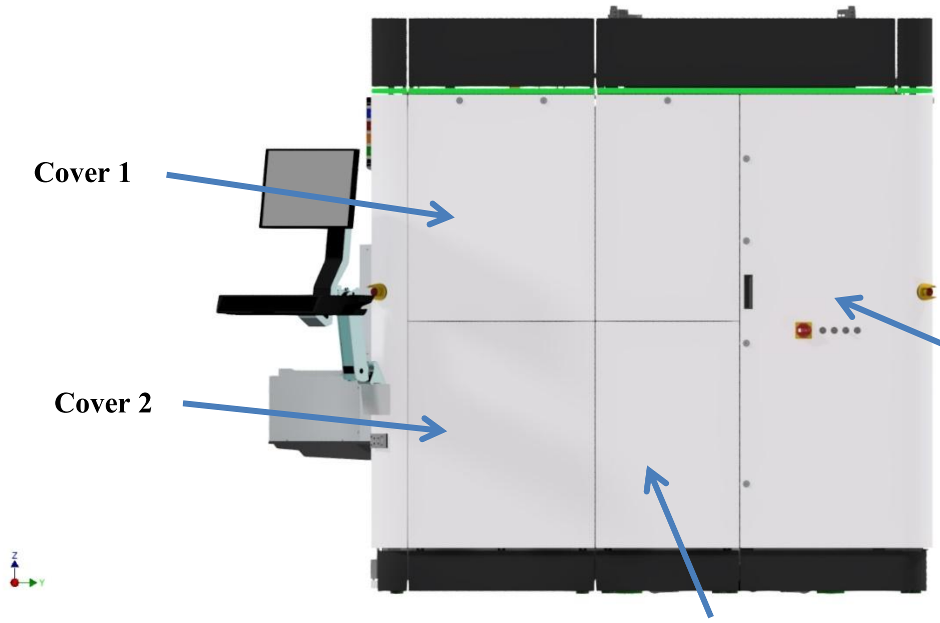
- Robot handling is stopped

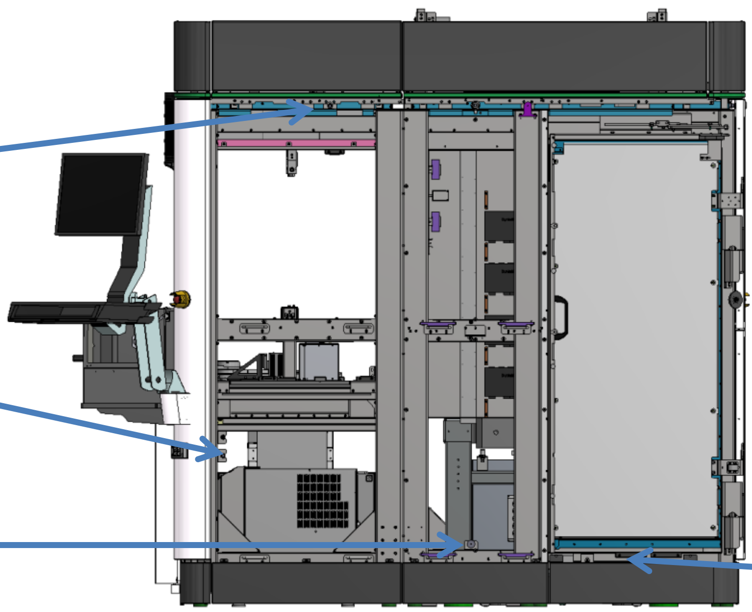
Documentation: Robot stops immediately → to be documented by checking “OK”. Otherwise “NOK”.

Arrangement of the door interlocks: See Figure 2.

Figure 2: DAVINCI – Location of the Door-Interlocks

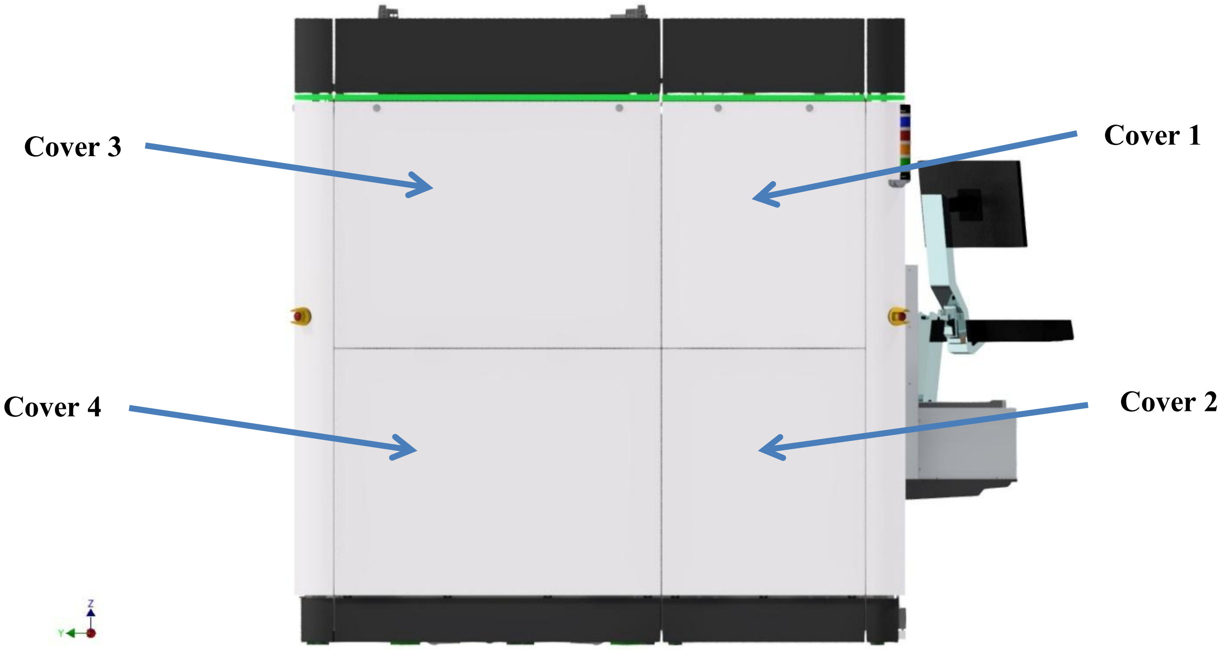
3.1 Robot and Scanning-Stage Door Interlock – Right Side

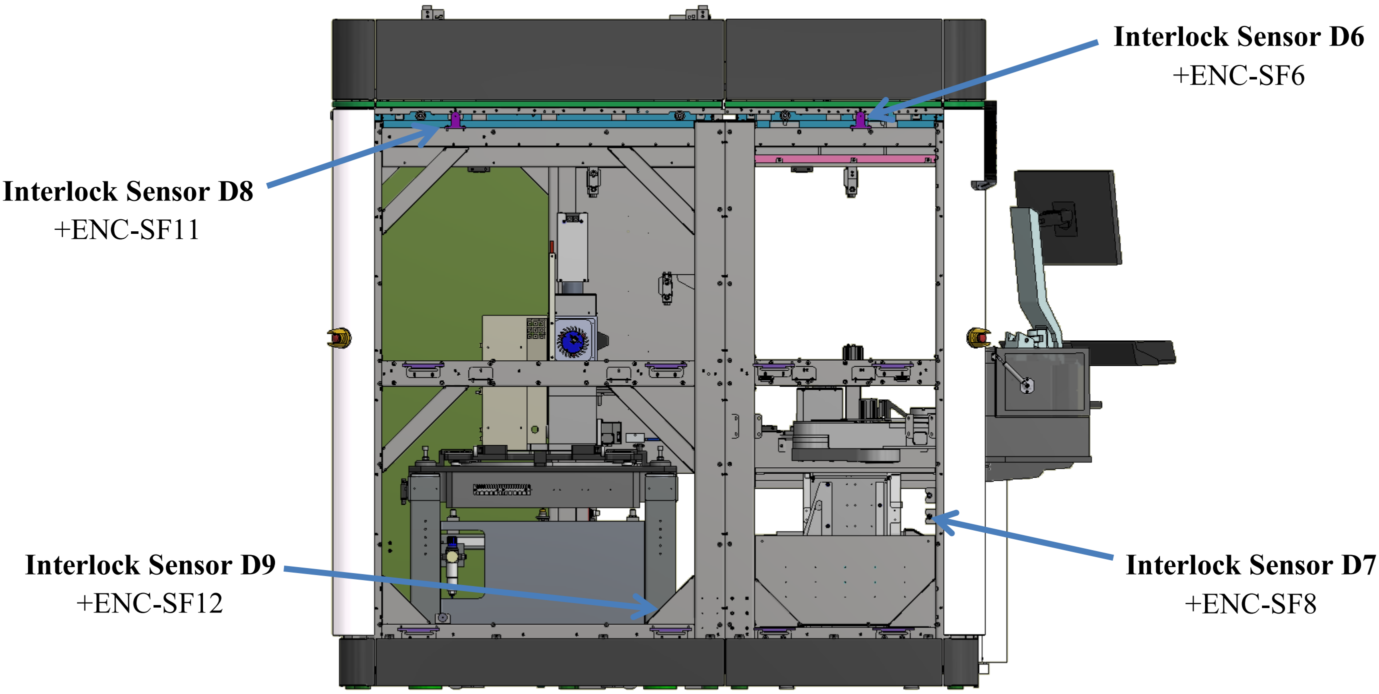




Door-Interlock #	EPLAN	Description	Action	Rob stops
D1	+ENC-SF5	Door Interlock EFEM top right	Remove Cover 1	
D2	+ENC-SF7	Door Interlock EFEM bottom right	Remove Cover 2	
D4	+ENC-SF9	Door Interlock MP side bottom right	Remove Cover 3	
D5	+ENC-SF10	Door Interlock MP Control Cabinet	Open door 1 (inner part)	

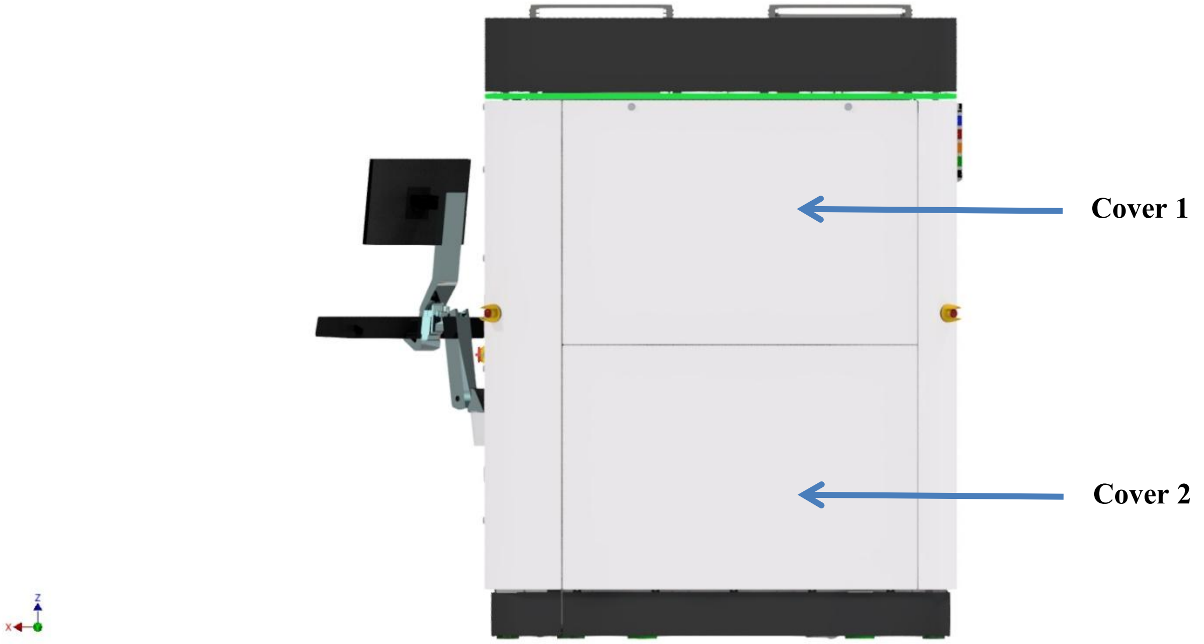
3.2 Robot and Scanning-Stage Door Interlock – Left Side

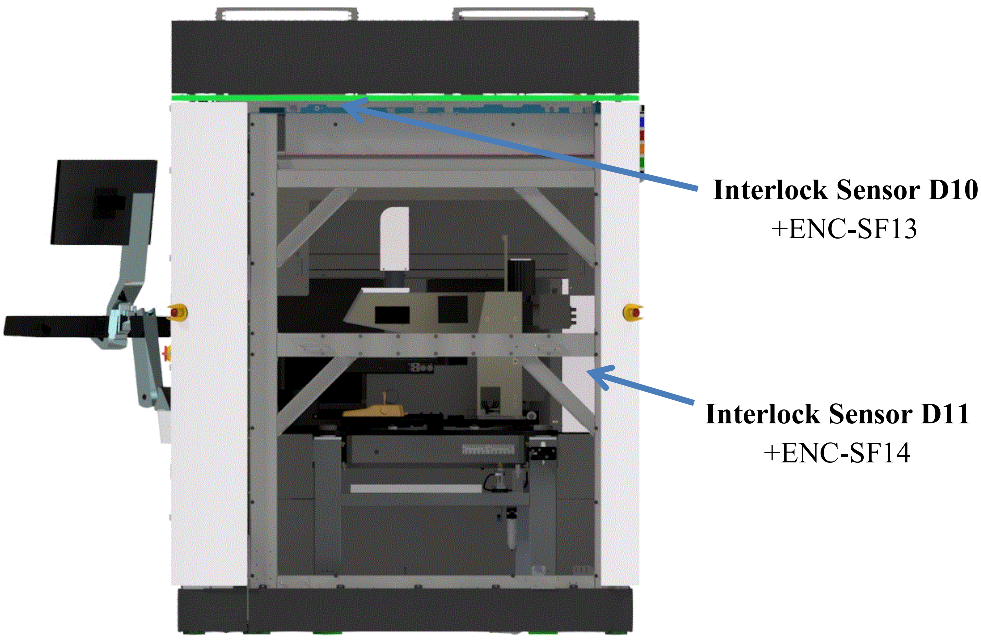




Door-Interlock #	EPLAN	Description	Action	Rob stops
D6	+ENC-SF6	Door Interlock EFEM top left	Remove Cover 1	
D7	+ENC-SF8	Door Interlock EFEM bottom left	Remove Cover 2	
D8	+ENC-SF11	Door Interlock MP side top left	Remove Cover 3	
D9	+ENC-SF12	Door Interlock MP side bottom left	Remove Cover 4	

3.3 Robot and Scanning-Stage Door Interlock – Rear





Door-Interlock #	EPLAN	Description	Action	Rob stops
D10	+ENC-SF13	Door Interlock MP rear top	Remove Cover 1	
D11	+ENC-SF14	Door Interlock MP rear bottom	Remove Cover 2	

3.4 Robot und Loadport Interlock

The correct function of the loadport interlocks is tested on the running system. It should be checked whether the robot stops when the pneumatic door is opened and the loading flap is opened simultaneously.

Action: Open the pneumatic loadport door. Open the loading flap.

Reaction: Emergency stop is executed:

- Robot handling is stopped

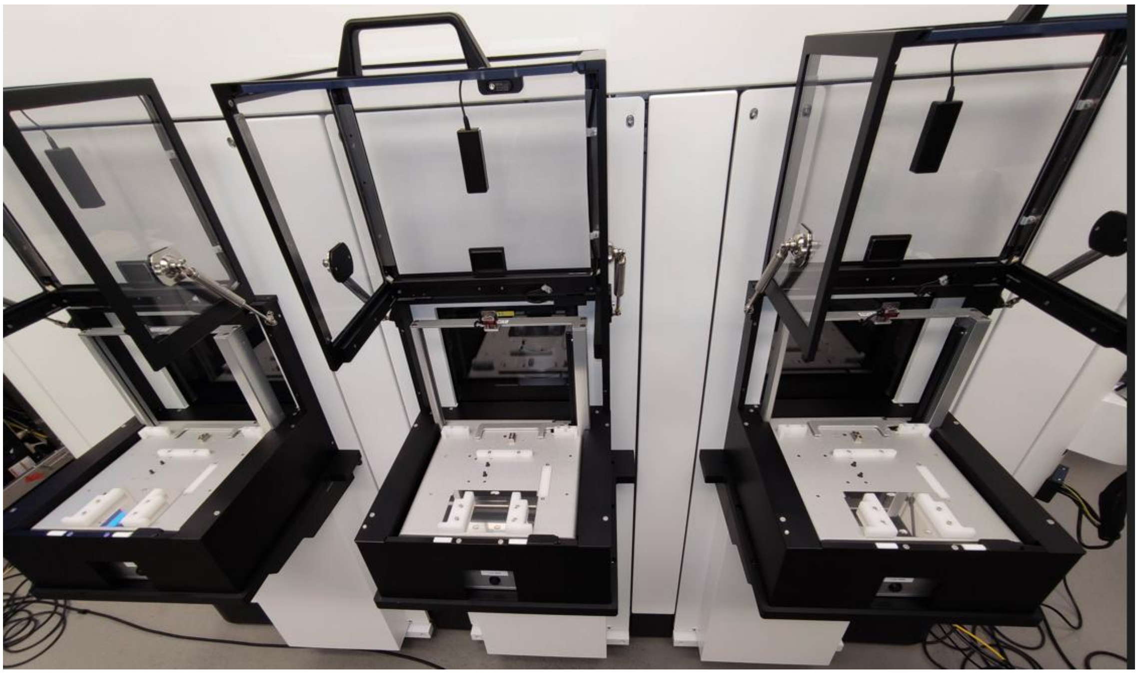


Figure 3: Loadports

Interlock #	EPLAN	Description	Action	Rob stops
LP1		Loadport1 (left) open pneumatic door	Open the loadport flap	
LP2		Loadport2 (middle) open pneumatic door	Open the loadport flap	
LP3		Loadport3 (right) open pneumatic door	Open the loadport flap	

4 Conducting the Tests

The tests are carried out based on the above specification as part of the acceptance at MueTec before delivery and documented in writing with date and signature.

4.1 Documentation of implementation and signatures

PLC -Version: Software Versions number und Date ______________________________

Safety-PLC: Project CRC __________________________

Test for System[µNummer] __________________ carried out on ________________________.

Remarks:

____

____

____

____

____

____

Test conducted by ____________________________________

Name in block letters

Place and Date ________________________________

____

Signature